

Chrony NTP Server & Client Configuration

Cluster: Virtual HPC Cluster

NTP Server: hpc-master (192.168.100.1)

NTP Clients: compute1, compute2, login, storage

Chrony Version: 4.5

NTP Server Configuration — hpc-master

Install Chrony

Chrony was installed on the master node using:

```
dnf install chrony -y
```

Configure /etc/chrony.conf

```
vi /etc/chrony.conf
```

Final configuration (/etc/chrony.conf):

```
# Use local system clock as time source

local stratum 10

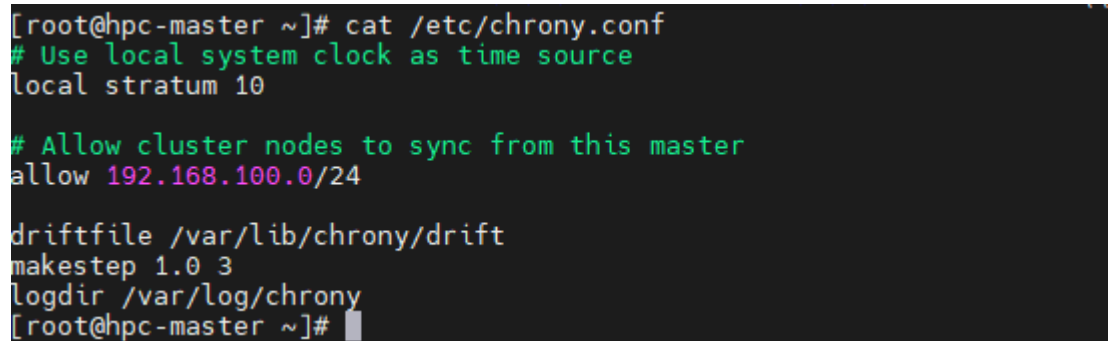
# Allow cluster nodes on this subnet to sync from this master

allow 192.168.100.0/24

driftfile /var/lib/chrony/drift

makestep 1.0 3

logdir /var/log/chrony
```

A terminal window screenshot showing the command 'cat /etc/chrony.conf' and its output. The output matches the configuration shown in the previous block, with some lines highlighted in green and pink in the original image.

```
[root@hpc-master ~]# cat /etc/chrony.conf
# Use local system clock as time source
local stratum 10

# Allow cluster nodes to sync from this master
allow 192.168.100.0/24

driftfile /var/lib/chrony/drift
makestep 1.0 3
logdir /var/log/chrony
[root@hpc-master ~]#
```

Explanation of each directive:

Directive	Meaning
local stratum 10	Use hpc-master's own system clock as the time source. Stratum 10 indicates a local reference.
allow 192.168.100.0/24	Allow all nodes on the cluster network to query this NTP server
driftfile	Records the clock drift rate so chronyd can compensate across restarts
makestep 1.0 3	Step the clock immediately if offset > 1 second during first 3 updates
logdir	Directory where chrony logs are stored

Start and Enable Chrony

```
# Start and enable chronyd
systemctl enable --now chronyd

# Restart to apply config changes
systemctl restart chronyd
```

Verify Master is Serving Time

Check tracking status:

```
chronyc tracking
```

Output:

```
Reference ID      : 7F7F0101 ( )
Stratum          : 10
Ref time (UTC)   : Mon Mar 02 05:59:07 2026
System time      : 0.000000000 seconds fast of NTP time
Leap status      : Normal
```

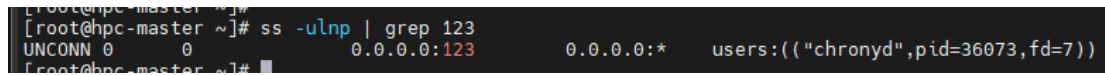
```
[root@hpc-master ~]# chronyc tracking
Reference ID      : 7F7F0101 ( )
Stratum          : 10
Ref time (UTC)   : Mon Mar 02 11:02:49 2026
System time      : 0.000000000 seconds fast of NTP time
Last offset      : +0.000000000 seconds
RMS offset       : 0.000000000 seconds
Frequency        : 0.000 ppm slow
Residual freq    : +0.000 ppm
Skew             : 0.000 ppm
Root delay       : 0.000000000 seconds
Root dispersion  : 0.000000000 seconds
Update interval  : 0.0 seconds
Leap status      : Normal
[root@hpc-master ~]#
```

Check NTP port is listening:

```
ss -ulnp | grep 123
```

Output:

```
UNCONN 0 0 0.0.0.0:123 0.0.0.0:*  
users: (("chronyd",pid=36073,fd=7))
```



```
[root@hpc-master ~]# ss -ulnp | grep 123  
UNCONN 0 0 0.0.0.0:123 0.0.0.0:* users: (("chronyd",pid=36073,fd=7))  
[root@hpc-master ~]#
```

- chronyd is listening on UDP port 123 on all interfaces
- Stratum 10 confirms local clock is active as time source
- Leap status Normal confirms clock is stable
- Sources list is empty — expected, as no upstream NTP server is used

NTP Client Configuration — All Cluster Nodes

All four client nodes (compute1, compute2, login, storage) were configured simultaneously using xCAT's `xdcp` and `xdsh` tools from hpc-master.

Verify Chrony is Installed on All Nodes

```
xdsh compute1,compute2,login,storage "dnf install chrony -y"
```

Output confirmed chrony 4.5 was already installed on all nodes:

```
compute1: Package chrony-4.5-1.el8.x86_64 is already installed.
compute2: Package chrony-4.5-1.el8.x86_64 is already installed.
login:    Package chrony-4.5-1.el8.x86_64 is already installed.
storage:  Package chrony-4.5-1.el8.x86_64 is already installed.
```

Create Client Configuration File on Master

```
cat > /tmp/chrony-client.conf << 'EOF'
# Sync time from hpc-master only
server 192.168.100.1 iburst

makestep 1.0 3
driftfile /var/lib/chrony/drift
logdir /var/log/chrony
EOF
```

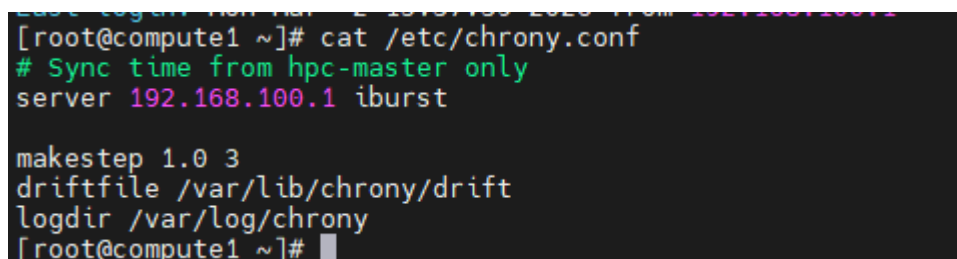
Explanation:

Directive	Meaning
server 192.168.100.1 iburst	Sync from hpc-master (cluster internal IP). <code>iburst</code> sends a burst of packets on startup for faster initial sync
makestep 1.0 3	Step the clock immediately if offset > 1 second during first 3 updates
driftfile	Stores clock drift information across reboots
logdir	Directory where chrony logs are stored

Push Configuration to All Client Nodes

```
xdcp compute1,compute2,login,storage /tmp/chrony-client.conf
/etc/chrony.conf
```

This command copies the client chrony config to `/etc/chrony.conf` on all four nodes simultaneously.



```
[root@compute1 ~]# cat /etc/chrony.conf
# Sync time from hpc-master only
server 192.168.100.1 iburst

makestep 1.0 3
driftfile /var/lib/chrony/drift
logdir /var/log/chrony
[root@compute1 ~]#
```

Enable and Restart Chrony on All Clients

```
xdsh compute1,compute2,login,storage "systemctl enable --now chronyd  
&& systemctl restart chronyd"
```

This ensures chronyd is enabled to start on boot and immediately applies the new configuration.

Verification — Client Nodes

Check NTP Source on All Clients

```
xdsh compute1,compute2,login,storage "chronyc sources"
```

Expected output on each node:

```
MS Name/IP address      Stratum Poll Reach LastRx Last sample
=====
^* 192.168.100.1         10      6   377      5   +0.1ms[+0.1ms] +/-
0.5ms
```

```
[root@hpc-master ~]#
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "chronyc sources"
compute1: MS Name/IP address      Stratum Poll Reach LastRx Last sample
compute1: =====
compute1: ^* hpc-master.hpc.local    10     10   377    504   +142us[ +112us] +/-  465us
storage: MS Name/IP address      Stratum Poll Reach LastRx Last sample
storage: =====
storage: ^* hpc-master.hpc.local    10     10   377     89    -29us[ -32us] +/-  486us
login: MS Name/IP address      Stratum Poll Reach LastRx Last sample
login: =====
login: ^* hpc-master.hpc.local    10     10   377    405   +195us[ +213us] +/-  641us
compute2: MS Name/IP address      Stratum Poll Reach LastRx Last sample
compute2: =====
compute2: ^* hpc-master.hpc.local    10     10   377    317   -582us[ -604us] +/- 1442us
[root@hpc-master ~]#
```

Check Tracking on All Clients

```
xdsh compute1,compute2,login,storage "chronyc tracking | grep -E
'Reference|Stratum|Leap'"
```

Expected output on each node:

```
Reference ID      : C0A86401 (192.168.100.1)
Stratum           : 11
Leap status       : Normal
```

```
[root@hpc-master ~]#
[root@hpc-master ~]# xdsh compute1,compute2,login,storage "chronyc tracking | grep -E 'Reference|Stratum|Leap'"
compute1: Reference ID      : C0A86401 (hpc-master.hpc.local)
compute1: Stratum           : 11
compute1: Leap status       : Normal
login: Reference ID        : C0A86401 (hpc-master.hpc.local)
login: Stratum             : 11
login: Leap status         : Normal
compute2: Reference ID      : C0A86401 (hpc-master.hpc.local)
compute2: Stratum           : 11
compute2: Leap status       : Normal
storage: Reference ID      : C0A86401 (hpc-master.hpc.local)
storage: Stratum           : 11
storage: Leap status       : Normal
[root@hpc-master ~]#
```

IMPORTANT COMMANDS

Purpose	Command
Check chrony service status	<code>systemctl status chronyd</code>
Check time sync status	<code>chronyc tracking</code>
Check NTP sources	<code>chronyc sources -v</code>
Force immediate time step	<code>chronyc makestep</code>
Check NTP port listening	<code>ss -ulnp grep 123</code>
Push config to all nodes	<code>xdcp <nodes> /tmp/chrony-client.conf /etc/chrony.conf</code>
Restart chrony on all nodes	<code>xdsh <nodes> "systemctl restart chronyd"</code>
Check sources on all nodes	<code>xdsh <nodes> "chronyc sources"</code>
Verify master allows clients	<code>chronyc accheck 192.168.100.13</code>